

ZCAMS

Zambia Customs Agent Management System

An end-to-end digital workflow for CFA onboarding, Bill of Lading review, Z-SAD issuance, GN 83-compliant invoicing, CapitalPay settlement, and cargo release.

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ZAFFA Clearing & Forwarding

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Proof-of-Concept overview

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ZCAMS is a working proof-of-concept that automates the day-to-day operations of a Clearing & Forwarding Agent (CFA) in Zambia. Built as a Plotly Dash web application with a SQLite back-end, it covers the full cargo-clearance lifecycle—from a CFA's first onboarding request through ZAFFA approval, Bill of Lading capture, Z-SAD issuance, GN 83-compliant invoice generation, CapitalPay payment settlement, and final cargo release.

The POC replaces a fragmented mix of paper forms, spreadsheets, email threads, and ad-hoc bank instructions with a single auditable system that enforces statutory fee rules, signs every invoice through CapitalPay, and shares each invoice with the importer over WhatsApp, SMS, and email automatically.

The Problem

Customs clearing in Zambia is a high-volume, high-risk process governed by Government Notice 83 (GN 83), which prescribes minimum agency fees by route, transport mode, and cargo category. Today the process is handled with manual tools that introduce five recurring problems:

- **Inconsistent fee enforcement.** Service-fee and full-settlement invoices are calculated by hand, so the GN 83 minimum, the 20% administrative fee, and the 16% VAT are frequently misapplied or omitted entirely.
- **Price undercutting.** Without a system-enforced floor, competing CFAs quote below the GN 83 statutory minimum to win work. This erodes margins for compliant agents, distorts the market, and leaves ZAFFA with no objective way to police the tariff its members are required to observe.
- **Late and unpredictable payments.** Importers receive invoices through informal channels, often without a working payment link, so settlement is delayed by days or weeks while the CFA chases reminders manually. Cargo sits at the border accruing demurrage and the agent finances the gap from working capital.
- **Untraceable Z-SAD usage.** Single-use Z-SAD numbers are recycled, lost, or issued against the wrong Bill of Lading, leaving no paper trail when ZRA queries a clearance.
- **Manual importer communications.** Invoices are emailed individually, payment links are pasted into chat apps, and cargo release confirmations are sent verbally, so disputes about what was sent, when, and to whom are unavoidable.

The cumulative effect is slow turnaround, revenue leakage from under-invoiced and underpriced jobs, working capital tied up in unpaid invoices, and exposure during ZRA audits.

ZCAMS turns the clearance workflow into a guided, rule-enforced pipeline. Each step writes to a SQLite store with audit trails, every invoice is cryptographically signed via CapitalPay before it is released, and every importer-facing message is dispatched through the platform so it can be replayed if a dispute arises.

How a clearance flows through ZCAMS

The CFA staff member proceeds through nine numbered stages:

- **1. CFA onboarding.** A clearing agent applies through a public form; ZAFFA approves or rejects the registration and login credentials are emailed automatically.
- **2. Bill of Lading capture.** Uploaded BL PDFs are OCR-extracted (OpenAI/Groq adapter) and persisted, or captured manually if OCR is unavailable.
- **3. Reviewed BL & Z-SAD.** Each BL is reviewed and assigned a single-use Z-SAD number with strict uniqueness; corrections require an explicit detach that retires the old number and cancels its invoice.
- **4. Invoice request.** The CFA chooses *Service Fee Only* or *Full Settlement*; ZCAMS calculates the GN 83 minimum, the 20% admin fee, and 16% VAT, ceiling each line.
- **5. CapitalPay signing.** The invoice is sent to CapitalPay for signing and the returned URN is stamped onto the PDF. Mock invoices are refused in production.
- **6. Importer share.** The signed PDF is dispatched to the importer over the selected channels: WhatsApp click-to-chat, SMS, and an email with the PDF attached via the Bird Reach API.
- **7. CapitalPay check-out.** A payment link is generated; settlement status is tracked against the invoice.
- **8. Cargo release.** Full Settlement auto-releases on payment; Service Fee Only enables a manual release once the bank confirms the importer payment.
- **9. Audit & communication.** Every action writes a notification record; the ZCAMS Chat module answers Gazette Notice questions for the CFA.

Outcome

Turnaround time drops from days to minutes because the system enforces the rules, signs the invoice, and notifies the importer in a single click. Because every invoice is generated from the GN 83 schedule and signed by CapitalPay, **price undercutting becomes impossible**—the statutory floor is the cheapest invoice the system will ever produce. And because the signed PDF ships with a live CapitalPay check-out link to WhatsApp, SMS, and email at the same moment it is created, importers **pay promptly** instead of waiting on manual reminders. Every Z-SAD, invoice, payment, and release is preserved with a timestamp, making ZRA audits a query rather than an investigation.

ZCAMS ships with fourteen modules accessible from the persistent sidebar. Each module is a self-contained Dash page that talks to the shared SQLite repository.

Login / Onboarding	Public CFA registration, ZAFFA approval, automated credential emails.
Dashboard	At-a-glance KPIs: active BLs, pending Z-SADs, outstanding invoices, recent payments.
BLs	Upload or manually capture Bills of Lading; OCR-assisted data extraction.
Reviewed BL	Issue and manage Z-SAD numbers; safe detach & reissue when corrections are needed.
Invoices	Browse the invoice register, filter by status, and re-download signed PDFs.
Check-out	CapitalPay payment links and live settlement status per invoice.
Contracts	Importer/CFA contract templates and execution records.
Certificates	Single-use certificate evidence linked to the underlying BL and Z-SAD.
Company Profile	CFA company details, banking information, and authorised signatories.
Notifications	Append-only audit feed of every state change with timestamp and actor.
Support	In-app ticket queue and contact channels for the CFA team.
ZCAMS Chat	Gazette Notice-aware FAQ assistant; optional local LLM (Qwen 2.5 0.5B).
GN 83 Schedule	Browsable reference of every GN 83 minimum-fee row used for invoice calculation.

Workflow & compliance

- End-to-end pipeline from onboarding to cargo release.
- Strict Z-SAD uniqueness with safe detach & reissue.
- Single-use BL number enforcement.
- Cancellation cascade: detaching a Z-SAD cancels its invoice.
- Manual release gate for Service Fee Only flows.
- Auto-release on Full Settlement after payment confirmation.

Invoicing & payments

- GN 83 minimum fee lookup by route, mode, and cargo category.
- Service Fee Only: 20% admin + 16% VAT (ceiled to whole cents).
- Full Settlement: GN 83 minimum + 20% admin + 16% VAT (ceiled).
- CapitalPay signing with refusal of mock URNs in production.
- PDF invoice generation via ReportLab; signed copies persisted.
- CapitalPay check-out link generation per invoice.

Importer communications

- WhatsApp click-to-chat link auto-opens in a new browser tab.
- SMS share for low-bandwidth recipients.
- Bird Reach API email with the signed PDF attached.
- Per-channel delivery status reported back to the CFA.
- Notifications table records every share attempt with timestamp.

Operational & audit

- SQLite store created automatically on first launch.
- Append-only notifications/audit feed.
- Server-side error capture surfaces failures inside the modal.
- File-based *invoice_flow.log* for deep diagnostics.
- GN 83 Schedule reference page for traceable fee lookups.
- ZCAMS Chat for in-app Gazette Notice questions.

Bills of Lading

- BL number must be unique within ZCAMS.
- Doc type, route, transport mode, and ZRA regime are mandatory.
- OCR-extracted fields are presented for human review before save.

Z-SAD numbers

- Each Z-SAD is single-use and globally unique.
- Detaching a Z-SAD retires the number and cancels its invoice.
- Z-SAD history is preserved on the reviewed BL record.

Invoices & payments

- GN 83 minimum fee is the floor for Full Settlement invoices.
- All money values are ceiled to two decimals to avoid rounding loss.
- Mock CapitalPay invoices (*CPAYMOCK...*) are refused unless `CAPITALPAY_MODE=mock` is set explicitly.
- Full Settlement auto-releases the cargo on successful settlement.
- Service Fee Only requires a manual release after bank confirmation.

Integrations

Integration	Purpose	Status
CapitalPay	Invoice signing, check-out links, settlement webhooks.	Live (real credentials)
Bird Reach API	Importer email with signed PDF attachment.	Live (real credentials)
OpenAI / Groq OCR	Bill of Lading data extraction.	Adapter — enable via .env
WhatsApp click-to-chat	Importer messaging via wa.me deep link.	Live (no API key required)
Gmail SMTP	Onboarding approval emails.	Optional fallback
HuggingFace Transformers	Local Qwen 2.5 0.5B for ZCAMS Chat.	Opt-in (<code>CHAT_MODEL_ENABLED=true</code>)

ZCAMS is deliberately built on a small, well-understood stack so it can be deployed by a single operator without a dedicated DevOps function.

Layer	Technology
Web framework	Plotly Dash 4.1 (Flask under the hood)
UI components	Dash Mantine Components, Dash Iconify, custom CSS theme
Server runtime	Python 3.13, Werkzeug 3.1 development server
Persistence	SQLite (bundled, file-based)
PDF generation	ReportLab 4.5
Document OCR	PyMuPDF, pdf2image, Pillow + OpenAI/Groq adapter
Payments	CapitalPay REST API
Email	Bird Reach API (Gmail SMTP fallback)
Optional LLM	HuggingFace Transformers + Qwen 2.5 0.5B
Testing	pytest 9, Dash test client

Getting Started

Run the POC

From the workspace root in PowerShell:

```
.\run-zcams.ps1
```

Then open:

```
http://127.0.0.1:8050/
```

Run the test suite

```
.\test-zcams.ps1
```

Covers routing, invoice flow, Z-SAD detach, GN 83 lookup, and the live CapitalPay integration where credentials are present.

Operational notes

- The SQLite database is created automatically at *cfa-dash/data/zcams.db* on first launch.
- Real CapitalPay signing is enforced; mock invoices are refused unless CAPITALPAY_MODE=mock is set explicitly.
- Every invoice action is logged to *cfa-dash/data/invoice_flow.log* for diagnostics.
- Server-side errors are surfaced inside the invoice modal so the CFA never has to guess why a request failed.

Status: the POC is feature-complete for the nine-stage clearance workflow and has been verified end-to-end against live CapitalPay signing and Bird Reach email delivery. The next milestone is hardening for multi-tenant deployment and wiring the OCR adapter to a production OpenAI/Groq workspace.